



Final Project– Proposal

Data Analytics - Microsoft Power BI Specialist

Student Name

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Instructor

Eng. Sherihan Ali

Initiative

DEPI

Group Code: MNF3_DAT2_S1

TemplateLAB



رواد مصر الرقمية

Project Description

This project analyzes and visualizes daily ridership data from the New York Metropolitan Transportation Authority (MTA) using Power BI.

The dataset includes information about various transportation modes — Subways, Buses, LIRR, Metro-North, Access-A-Ride, Bridges & Tunnels, and Staten Island Railway — and compares current ridership with pre-COVID levels.

The objective is to build an interactive Power BI dashboard that reveals usage trends, recovery rates, and behavioral changes to support transportation planning and decision-making.

Group Members & Roles

Member Name	Role	Responsibility
Omar Hamed Ali	Team Leader / Data Analyst	Leads the team, manages the project timeline, and handles data cleaning, data modeling, and DAX calculations.
Soliman Yehia Soliman	Data Collection Specialist	Responsible for gathering the dataset and identifying all relevant data sources for analysis.
Mohamed Ahmed Mohamed AlFakharany	Data Engineer	Performs data modeling and data cleaning to ensure data quality and integrity.
Samir Mahmoud Samir	Data Exploration Analyst	Conducts exploratory data analysis (EDA) to uncover initial trends, anomalies, and key observations.
Ahmed Nasr Abdel Motey Mesalem	Dashboard Developer	Designs and builds the Power BI dashboard, ensuring usability, interactivity, and accurate visual representation.
Ahmed Hamdy	Reporting & Presentation Lead	Prepares the final project report, documentation, and delivers the team presentation.

Team Leader : Omar Hamed Ali

Objectives:

- Measure how MTA ridership recovered after COVID-19 across all services.
- Analyze weekday, weekend, and holiday travel behavior, including the **“Friday Effect.”**
- Compare recovery rates between subways, buses, rail, and paratransit services.
- Visualize trends and create actionable insights to support resource allocation.
- Evaluate financial impacts and propose strategies for better operational planning.

Tools & Technologies:

- Power BI – Data modeling, DAX calculations, dashboard visualization.
- Microsoft Excel / CSV – Data storage and preprocessing.
- DAX Language – Measures and calculated columns for KPIs.
- PowerPoint / Canva/ Gamma.ai– Report design and presentation.

Milestones & Deadlines:

Milestone	Description
Week 1 – Data Research & Understanding	Conduct background research on MTA ridership, understand dataset fields, and define project goals.
Week 2 – Data Collection	Gather the required datasets from official MTA sources and verify data accuracy and completeness.
Week 3 – Data Cleaning	Clean and preprocess the data (handle missing values, fix inconsistencies, and standardize formats).
Week 4 – Data Modeling	Build data model relationships, define schema, and prepare data for Power BI integration.
Week 5 – Calculation	Create calculated columns and DAX measures for KPIs and insights.
Week 6 – Visualization & Dashboard	Design and develop the Power BI dashboard, ensuring interactivity and clarity.
Week 7 – Report Writing	Document analysis results, findings, and recommendations in a clear report format.
Week 8 – Presentation & Final Submission	Prepare and deliver the final presentation, submit the dashboard and project report.

KPIs (Key Performance Indicators):

#	KPI Name	What It Means / Purpose	Formula / Calculation
1	Total Ridership	Measures the total number of passengers using all MTA services daily.	Sum of ridership across all modes.
2	Ridership Recovery %	Shows how current ridership compares to pre-COVID (2019) levels.	$(\text{Current Ridership} \div \text{2019 Average}) \times 100.$
3	Mode Share %	Indicates the proportion of each transport mode (subway, bus, rail etc.) in total ridership.	$(\text{Ridership of mode} \div \text{Total Ridership}) \times 100.$
4	Weekday vs Weekend Ratio	Compares weekday ridership to weekend to reveal work vs leisure travel behavior.	$(\text{Average Weekday Ridership} \div \text{Average Weekend Ridership}).$
5	Friday Effect Gap	Quantifies the drop in ridership on Fridays compared to Tuesday–Thursday levels.	$((\text{Tue–Thu Average} - \text{Friday}) \div \text{Tue–Thu Average}) \times 100.$
6	Traffic Recovery % (Bridges & Tunnels)	Measures on how vehicular traffic rebounded relative to pre-COVID levels.	$(\text{Current Traffic} \div \text{2019 Average}) \times 100.$
7	Revenue Impact	Estimates fare revenue change resulting from ridership variation.	$\text{Ridership} \times \text{Average Fare}.$
8	Data Quality Score (Accuracy & Completeness)	Evaluates the cleanliness and accuracy of data after preprocessing.	$\% \text{ of valid records after cleaning} = [(\text{Total} - \text{Errors}) / \text{Total}] \times 100.$
9	Visualization Effectiveness (Usability Score)	Assesses dashboard clarity and user experience based on peer feedback.	$\text{Positive Feedback} / \text{Total Feedback} \times 100.$